

You can stretch or compact most of the materials including polysilicon, diffusion and metal. However, components like transistors and contacts can only be moved, their shapes cannot be changed.

Technology file

Magic is a technology-independent layout editor where all the technology-specific information is contained in a technology file. This file includes information about layer types used, electrical connectivity between types, design rules, rules for mask generation and rules for extracting netlists for circuit simulation.

Since there is different technology file for every fabrication process, you can either build your own or re-write the existing one for a new fabrication process. You can download the standard technology file using FTP server at MOSIS foundation. You can select the symbolic link 'Current' for the latest version at the following URL:

<ftp://ftp.mosis.edu/pub/sondeen/magic/new/beta/current.tar.gz>

Why Magic

Magic is different from the other layout editors available. It is not just a colour-painting tool, it understands geometrical design rules, transistors, connectivity and routing. It is a very handy tool for easy bug fixing and experimentation with alternative designs to enhance the performance of a circuit.

One of its major features, which the other similar tools lack, is routing. Routing is the most tedious and error-prone task, but Magic has made it easy with its interactive routing tools. Another problem with the other tools is their inflexibility and difficulty to experiment with them.

But Magic's built-in knowledge of design rules and interactive feature of continuous design-rule check with an in-built hierarchical circuit extractor makes this tool highly flexible and easy to use. Also, its plowing feature ena-

bles students to experiment and have better intuition on designing chips and fix bugs easily.

Support

Magic is an open source tool available at <http://opencircuitdesign.com/magic/>

You can download the latest version of the tool and find the related tutorials on this website. If you are trying to learn about the system, you can start with the various tutorials available on the website. Also, there is a set of manuals for the system maintainers that can help in creating new technologies.

While running Magic, you can get help from the tool itself. Its help command will provide you the command's syntax with a brief description of the command. For other queries and support, you can write to Magic authors at magic-dev@cs.cornell.edu ●

The author is a technical journalist at EFY

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For any queries, please contact our team at efyengq@efy.in OR +91-11-26810601